

A Private AI Assistant For The Lab

What we can build, how fast it will feel, and what it will cost

`libr-local-llm`

The idea, in one sentence

Put an AI assistant **on the lab's own computers**, so several people can use it at once — and so it is allowed to read patient data, which no outside service ever will be.

Today

You use Claude. It is excellent and fast. But every word you send leaves the building, so it can never touch a patient recording or transcript.

What this would add

An assistant living on our own machines. A bit slower. Not quite as clever. But it can look at the actual data, because nothing ever leaves.

This is **not** a replacement for Claude. It is a second tool that can go where Claude is not allowed to go.

First, the one idea everything else depends on

An AI model is basically a gigantic pile of numbers. Ours would be a very big pile: **744 billion of them**. Far too big to keep somewhere fast and convenient.

Think of it as a library. The model is a library with about **19,000 books**. To produce even one word of an answer, it does not read the whole library — it fetches a small handful of books, reads them, and puts them back.

On the desk
tiny, instant

a few books

On the shelves
big, quick

all 19,000 fit here

In the basement
huge, slow

we avoid this

Good news, and the reason this project is possible at all: our computers have enough shelf space for *the entire library*. The assistant never goes to the basement while answering you.

Why it is slow: 600 trips to the shelf, per word

Here is the whole speed problem, and there is no trick that makes it go away.



$75 \text{ floors} \times 8 \text{ books} = \mathbf{600 \text{ book-fetches for every single word.}}$

Everything about the design comes from trying to make those 600 trips faster. We can shorten the walk. We can put the most-used books nearer. We **cannot** avoid the trips — that is simply how this kind of model works.

And the fetching is only *half* the work. The other half is the thinking it does once it has the books. Even if we made the fetching instant, we would only be about **twice** as fast. Worth knowing before anyone promises ten times.

The obvious fix that does not work

The obvious idea: if eight people ask at the same time, fetch everyone's books in *one* trip. Serve eight for the price of one. It fails, and the reason is simple:

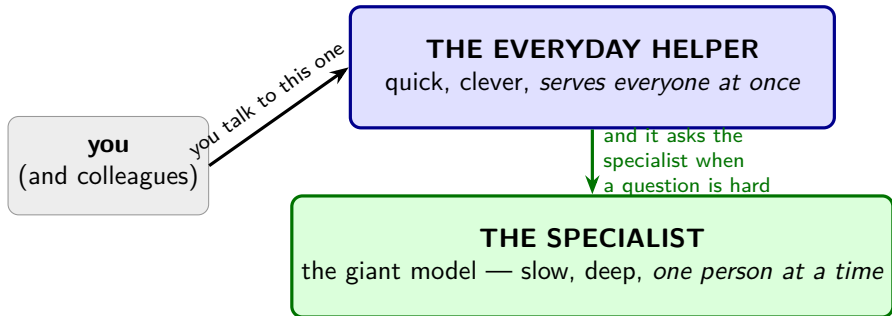
Eight people want eight *different* sets of books. One trip for 8 books becomes one trip for about **57** — and you are still only serving eight people.

People at once	Books to fetch	
1	8	
4	33	barely better than doing them one by one
8	57	

This was actually tested, not guessed. With eight people the group finished no faster in total — and each person waited **about five times longer** than going alone.

The conclusion: a giant model cannot serve a crowd quickly. For a crowd you need a **smaller** model. That is why we build more than one.

So: three helpers, not one



Like a good office. You talk to the capable colleague who is always there. When something genuinely difficult comes up, *they* ask the world expert down the hall and come back with the answer.

A **third** helper runs overnight, grinding through big piles of paperwork — scoring thousands of transcripts. Nobody talks to it.

Helper 1 — the everyday one

How big	about a third the size of the specialist — still very large
Where it lives	our one machine with four graphics cards
How many people	everybody, at the same time
How fast	roughly a page of text every 20–30 seconds
What it is for	almost everything you do all day

Why this one can serve a crowd and the specialist cannot: it is small enough to live entirely on the graphics cards, which are enormously faster than the shelves. The book-fetching stops being the problem, so serving several people at once genuinely does work here.

We have not picked exactly which model yet — that is the first real job. It has to fit on the four cards, be good at code, and be able to use tools. **It is not the small model that was tried and rejected earlier:** this one has roughly eight times as many numbers in it and runs on much better software.

Helper 2 — the specialist

How big	744 billion — the biggest thing we can run at all
Where it lives	one ordinary machine, using most of its memory
How many people	one at a time. Others wait in line
How fast	roughly a page of text per minute
What it is for	the hard question you would otherwise be stuck on

One catch worth planning around. Waking it up means reading 372 gigabytes off the shared storage. That takes about **27 minutes**. So we wake it in the morning and leave it awake, rather than starting it on demand.

It is slow because it is enormous, and it is worth having *because* it is enormous. A minute of waiting for a genuinely hard answer is a good trade. A minute of waiting to rename a variable is not — which is exactly why helper 1 exists.

How good is the specialist, really?

The honest answer depends entirely on *what kind* of question you ask.

What you ask	Versus Claude	How long
“Here is a function and a bug report — what is wrong?”	genuinely close	1–4 minutes
“Which of these two designs is better, and why?”	genuinely close	1–4 minutes
“Read these eight files and find the inconsistency”	not close	several minutes before it even starts
“Keep working on this with me for an hour”	don’t	see the next slide

One hard question, with everything it needs already in front of it. That is what the specialist is for, and there it is worth the wait. When the job becomes “hold a lot in your head and notice something subtle”, the gap opens wide.

An awkward twist: the harder the question, the more it thinks before answering — and thinking is what it is slow at. The questions most worth asking are the ones you wait longest for.

“Can I just write down what I want and let it code?”

Yes — with the everyday helper, and with limits worth knowing first.

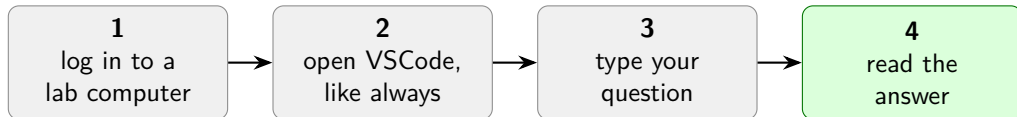
A moderate coding job, start to finish	How long	Sensible?
Claude, today	3–5 minutes	yes
the everyday helper	6–12 minutes	yes
the specialist	about 30 minutes	no — leave it alone

Two to four times slower than Claude for the same job. Write the description, start it, go and do something else for ten minutes. A real workflow, not a demo.

The catch is not about speed. To edit your files the model fills in a little form for each action, and because we run a *compressed* copy it **sometimes fills the form in wrong**. There is a built-in repair for the simple forms — open a file, run a command — but **not for the complicated ones**, and “change this line to that line” is a complicated one.

Start with small, clearly described jobs and check the result. How often it fills the form in wrong is one of the first things we measure — if it is often, “hand it a spec” is a demo, and we want to know that early.

What it looks like to use

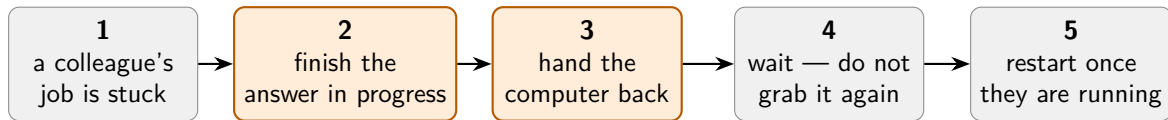


That is the whole thing. The same terminal window you already use. The same way of asking. You would not be able to tell from the screen that the answer came from down the hall instead of from California.

<code>fleet code</code>	start working. Talks to the everyday helper
<code>fleet ask "..."</code>	send one hard question to the specialist and wait
<code>fleet status</code>	is it running? is anyone queued?
<code>fleet down</code>	stop everything

Being a good neighbour

These are shared computers. Other people in the lab are running their own work on them, and our assistant would be sitting on a fair chunk of them all day.



Step 4 is the one that matters. Without it the system hands a computer back, notices a second later that it is short of computers, and grabs the same one straight back — beating the very colleague it was trying to help. It would look rude while running code written to be polite.

It switches itself off when nobody has used it for a few hours, and back on when someone does. It restarts itself if it crashes. Nobody babysits it.

How we will know whether it is any good

Speed we can measure. **Whether it is *smart* enough is harder** — and we have an unusually good way to test it.

Use this project as the exam. Planning this system turned up three real discoveries about our own hardware that nobody had noticed. We know the answers now. Hand the local assistant the same starting point and see how many it finds on its own.

Would we bet on it?	The discovery it has to make
maybe probably probably not	one of our job queues can silently freeze the assistant mid-answer asking for memory quietly charges us for twice the computers why serving eight people at once cannot work

Grade this hardest: does it *hold its ground* when told it is wrong? One that folds and invents a better-sounding answer is worse than one that never spotted the problem.

What to expect: six people log in on Monday morning

Question	Honest answer
Can six people use it at once?	Yes. All six talk to the everyday helper, all at the same time, and it is built for exactly that.
How fast for each of them?	A page of text in 20–30 seconds. It does not noticeably slow down as colleagues join.
How does that compare to Claude?	Claude does that page in about 9 seconds . So: two to three times slower. Noticeable. Not painful.
What if two people need the specialist?	They queue . It serves one at a time, about a minute a page, so the second person waits about two minutes.
First thing Monday?	The specialist needs 27 minutes to wake up. We wake it before people arrive.
Will it be as clever as Claude?	For everyday work, close. For long, tangled problems across many files — no , and we should not pretend otherwise.

Everything above is an estimate built from real measurements on similar machines. The first thing we build is the test that replaces these numbers with ours.

The bottom line

**Six people. Their own terminal. A page of text every 20–30 seconds.
A genius down the hall for the hard ones.
And all of it allowed to read the data.**

What we are promising

- Everyone can use it at once
- Two to three times slower than Claude
- Good enough for everyday work
- Nothing ever leaves the building
- Nobody has to babysit it

What we are not promising

- It will not match Claude on hard problems
- The specialist serves one person at a time
- It uses a real share of the lab's computers
- Every number here still needs confirming

Slower. Not as clever. Allowed in the room where the data is.

That last one is the only reason to build it — and it is enough.